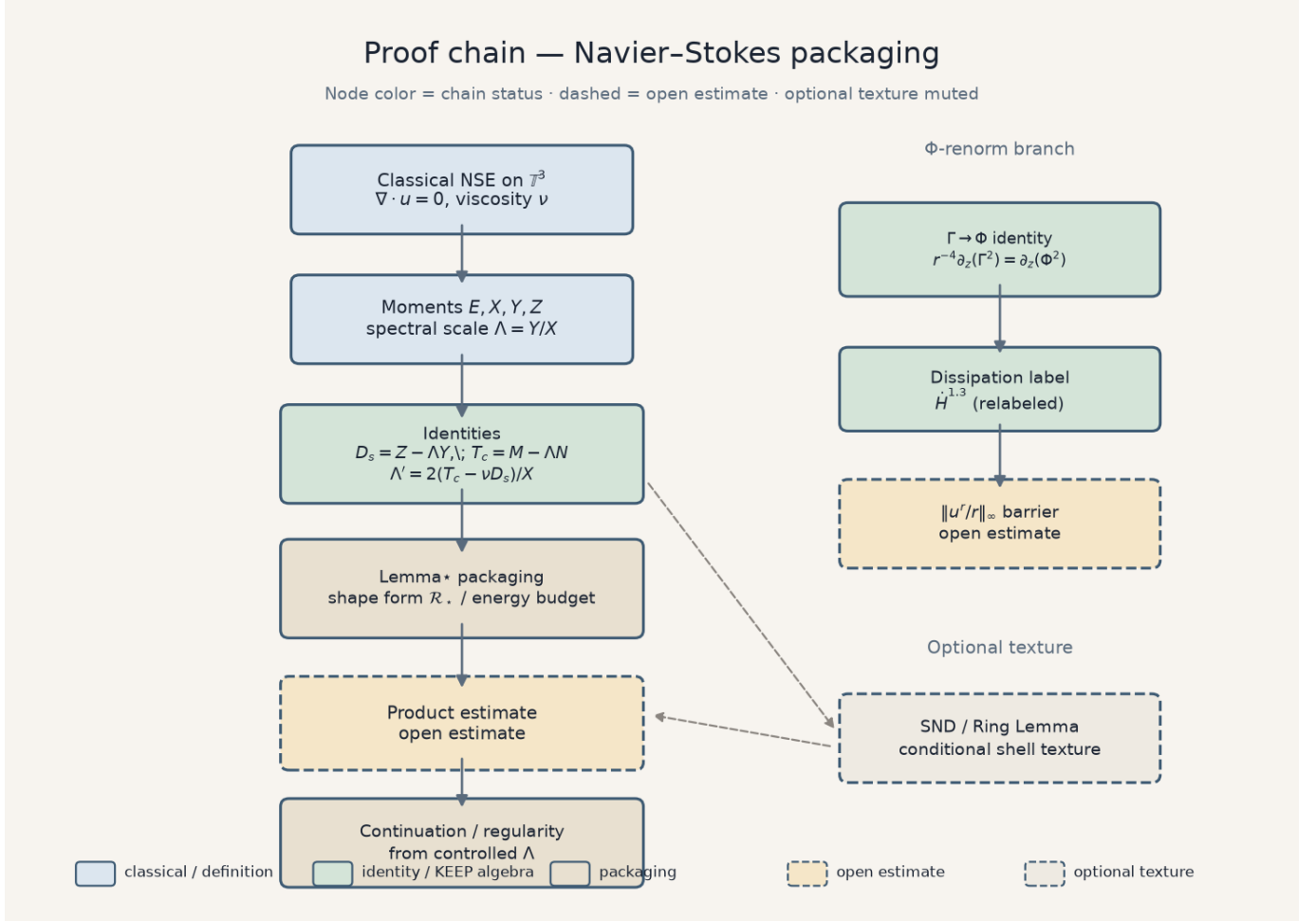


Proof chain (figures)

Navier–Stokes packaging · spectral moments · Lemma★ · open estimates

Main trunk on $\mathbb{T}^3$; Φ -renorm as a separate branch; SND as optional texture. Node colors mark chain status. Dashed nodes are open estimates.



Definitions. $E = \|v\|_2^2$, $X = \|A^{1/2}v\|_2^2$, $Y = \|Av\|_2^2$, $Z = \|A^{3/2}v\|_2^2$, $\Lambda = Y/X$, $D_s = Z - \Lambda Y$, $T_c = -\langle B(v, v), A(A - \Lambda)v \rangle$.

Lemma★ (shape). $(T_c)_+^2 \leq C_{\text{geom}} D_s E Y$, or $\mathcal{R}_* = (T_c)_+^2 / (D_s E Y)$ uniformly bounded.

Open estimate. Product-class control of $|T_c|$ from Leray-scale quantities (HH-channel bottleneck in the Bony diagnostic).

Φ -renorm. $\Gamma = ru_\theta$, $\Phi = \Gamma/r^2$; identity $r^{-4}\partial_z(\Gamma^2) = \partial_z(\Phi^2)$; dissipation label $\dot{H}^{1,3}$; barrier $\|u^r/r\|_\infty$ open.

Companion: docs/ns-review/PROOF-CHAIN-CLEAN.md. Figures: docs/ns-review/visual-journey/.